MONASH University

ETC4500/ETC5450 Advanced R programming

Week 5: Debugging

Outline

1. Debugging strategy
2. Minimal reproducible examples
3. Debugging tools
4. Styling

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1 Debugging strategy

2 Minimal reproducible examples

3 Debugging tools

4 Styling

What's a bug?

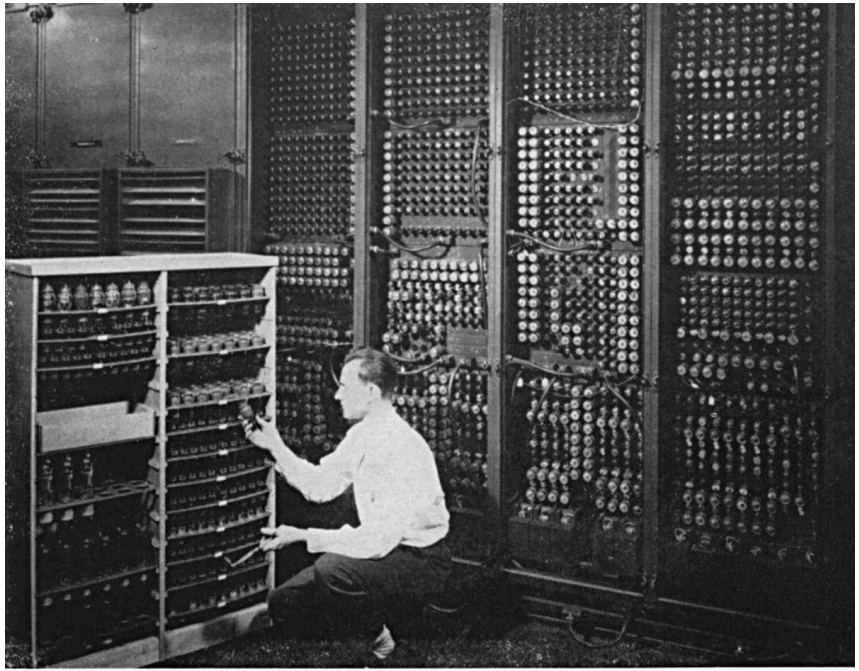
An incorrect, unexpected, or unintended behaviour of code.

Why do we call it a bug?

Why not a mistake? A glitch? An oopsie-daisy?

What's a bug?

On September 9, 1947, a real moth was found causing a malfunction in the Harvard Mark II computer. This incident was recorded in the logbook with the note "First actual case of bug being found."



Replacing a bad tube meant checking among ENIAC's 19,000 possibilities.

Overall debugging strategy

Ask for help

- Ask an LLM (OpenAI, Claude, ...)
- Ask a search engine (Google, Bing, DuckDuckGo, ...)
- Ask the community (Stack Overflow / Posit Community, ...)

Fix it yourself

- Update your software / R packages
- Create a minimal reproducible example
- Explore code to find where the error is
- Create a unit tests with expected behaviour
- Fix and test it

Asking for help

To get useful help, it is important that you ask a **good question**. Consider answering these two equivalent questions, which is easier to understand and why?

Asking for help

! urgent help needed with assignment error

My code doesn't work. Please help i need it working for my assignment asap!

data <- read.csv("C://Users/James/Downloads/project-a9j-2020a/files/survey_data.csv") data %>% filter(y == "A") %>% ggplot(aes(y = y, x = temperature)) + geom_line()

Asking for help

Error with dplyr `filter()`: "object not found"

I'm trying to filter a dataset in `dplyr`, but I'm getting an error that I don't understand. Here's my code and error message:

```
survey <- data.frame(x = c(1, 2, 3), y = c("A", "B", "C"))
survey %>% filter(y == "A")
```

Error: `Error in filter(y == "A") : object 'y' not found`

I expected it to return rows where `y` is `"A"`. How should I fix this?

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Minimal reproducible examples

A minimal reproducible example (MRE) is essential for effectively communicating problems with code.

The process of creating a MRE might also help you resolve the problem yourself!

Minimal reproducible examples

Minimal: Minimising code and data makes it easier to find the problem.

- Remove unnecessary code
- Prefer built-in datasets or small example datasets.
- Remove unused packages or files irrelevant to the problem.

Minimal reproducible examples

Reproducible: Ensure that others can run your code and see the same problem.

- If external packages are needed, include loading the packages in your MRE.
- If you can't use built-in datasets, provide a minimal dataset with `data.frame()` or `dput()`.
- If your problem includes randomisation, include `set.seed()` with appropriate seed.

Minimal reproducible examples

Examples: Provide a clear example of the problem.

- Explain what you expect versus what happens.
- Add code comments to highlight your intention and the problem.

reprex

The **reprex** package helps create *minimal reproducible examples*.

- Results are saved to clipboard in form that can be pasted into a GitHub issue, Stack Overflow question, or email.
- `reprex::reprex()`: takes R code and outputs it in a markdown format.
- Append session info with `reprex(..., session_info = TRUE)`.

Exercise: MRE

Create a Minimal Reproducible Example (MRE) for this code:

```
library(tidyverse)
library(rainbow)

survey_data <- read.csv("https://arp.numbat.space/week5/survey_data.
csv")

survey_data |>
  select(-RespondentID) |>
  group_by(Gender) |>
  count(Satisfaction)
```

https://arp.numbat.space/week5/survey_dplyr_bug.R

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The debugging workflow

1. Create a reprex that demonstrates the problem as a comment in the issue.
2. Fix the problem in the package code.
3. Add a comment to the issue explaining the bug and the fix, including a link to the commit containing the fix.
4. Add unit test(s) to the package that confirms the problem is fixed.
5. Close the issue.

Debugging tools in R

- `traceback`: prints out function call stack after error occurs; does nothing if no error.
- `debug`: flags function for “debug” mode which allows you to step through execution of function one line at a time.
- `undebug`: removes the “debug” flag from a function.
- `browser`: pauses execution of a function and puts the function in debug mode.
- `trace`: insert code into a function at a specific line number.
- `untrace`: removes the code inserted by `trace`.
- `recover`: allows you to modify the error behaviour so that you can browse the function call stack after an error occurs.

Traceback

```r
f <- function(a) g(a)
g <- function(b) h(b)
h <- function(c) i(c)
i <- function(d) {
  if (!is.numeric(d)) {
    stop("`d` must be numeric", call. = FALSE)
  }
  d + 10
}
```

```
> f("a")
Error:
! `d` must be numeric
▶ Show Traceback ✧Fix ✧Explain
```

```
> f("a")
Error:
! `d` must be numeric
▼ Hide Traceback ✧Fix ✧Explain

    1. └─global f("a")
    2.   └─global g(a)
    3.     └─global h(b)
    4.       └─global i(c)
```

Exercise

Exercise: traceback

https://arp.numbat.space/week5/01-traceback.R

Interactive debugging

- Using `browser()`

```
i <- function(d) {
  browser()
  if (!is.numeric(d)) stop("`d` must be numeric", call. = FALSE)
  d + 10
}
```

- Set breakpoints in Positron or RStudio by clicking to left of line number, or pressing `Shift+F9`.
- `options(error = browser)`

Exercise

Exercise: browser

`https://arp.numbat.space/week5/02-browser.R`

Interactive debugging

Debugging commands:

1. `n`: Next line (step over).
2. `s`: Step into function.
3. `c`: Continue to next breakpoint.
4. `f`: Finish the current function.
5. `Q`: Quit debugging.
6. `where`: Show the call stack.
7. `help`: Help with these debugging commands.

Interactive debugging

- `debug()` : inserts a `browser()` statement at start of function.
- `undebug()` : removes `browser()` statement.
- `debugonce()` : same as `debug()`, but removes `browser()` after first run.

Exercise

Exercise: debug

https://arp.numbat.space/week5/03-debug.R

Common error messages

- could not find function `"xxxx"`
- object `xxxx` not found
- missing value where `TRUE` / `FALSE` needed
- attempt to apply non-function
- undefined columns selected
- subscript out of bounds
- object of type ‘closure’ is not subsettable
- `$` operator is invalid for atomic vectors
- list object cannot be coerced to type ‘double’
- arguments imply differing number of rows
- non-numeric argument to binary operator

Exercise

Exercise: errors

https://arp.numbat.space/week5/04-errors.R

Common warning messages

- NAs introduced by coercion
- replacement has `xx` rows to replace `yy` rows
- number of items to replace is not a multiple of replacement length
- the condition has length > 1 and only the first element will be used
- longer object length is not a multiple of shorter object length
- package is not available for R version `xx`

Common warning messages

- NAs introduced by coercion
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Tip: turn warnings into errors with `options(warn = 2)` to use `traceback()`

Exercises

Exercise: warnings

https://arp.numbat.space/week5/05-warnings.R

Exercises

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https://arp.numbat.space/week5/05-warnings.R

Exercise: debug

https://arp.numbat.space/week5/06-debug.R

Exercises

Exercise: warnings

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Exercise: debug

https://arp.numbat.space/week5/06-debug.R

Exercise: debug

https://arp.numbat.space/week5/07-debug.R

Non-interactive debugging

- Necessary for debugging code that runs in a non-interactive environment.
- Is the global environment different? Have you loaded different packages? Are objects left from previous sessions causing differences?
- Is the working directory different?
- Is the `PATH` environment variable, which determines where external commands (like `git`) are found, different?
- Is the `R_LIBS` environment variable, which determines where `library()` looks for packages, different?

Non-interactive debugging

- `dump.frame()` saves state of R session to file.

```r
dump_and_quit <- function() {
  # Save debugging info to file last.dump.rda
  dump.frames(to.file = TRUE)
  # Quit R with error status
  q(status = 1)
}
options(error = dump_and_quit)
# In a later interactive session ----
load("last.dump.rda")
debugger()
```

- Last resort: `print()`: slow and primitive.

Other tricks

- `sink()` : capture output to file.
- `rlang::with_abort()` : turn messages into errors.
- If R or RStudio crashes, it is probably a bug in compiled code.
- Post minimal reproducible example to Posit Community or Stack Overflow.

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Style guides

Tidyverse

https://style.tidyverse.org/

Google

https://google.github.io/styleguide/Rguide.html

Indentation

- Use **2 spaces** per indentation level.
- Add spaces around operators: `x <- y + z`.

Naming (functions, arguments, objects)

Be brief but descriptive with object names.

Use a consistent naming convention:

- camelCase
- snake_case
- PascalCase

Design

- **Modularity**: Create re-usable parts for maintainability and scalability.
- **Simplicity**: Keep the interface intuitive and easy to use with straightforward interactions.
- **Flexibility**: Allow adaptability to different use cases and user preferences.
- **Feedback**: Provide clear and timely feedback to inform users of actions, errors, and system states.

Automatic styling

- styler: `https://styler.r-lib.org/`
- air: `https://posit-dev.github.io/air/`

These can be configured to automatically style your code when you save.

You can also check your code for common problems with:

- `lintr`
- `jarl`